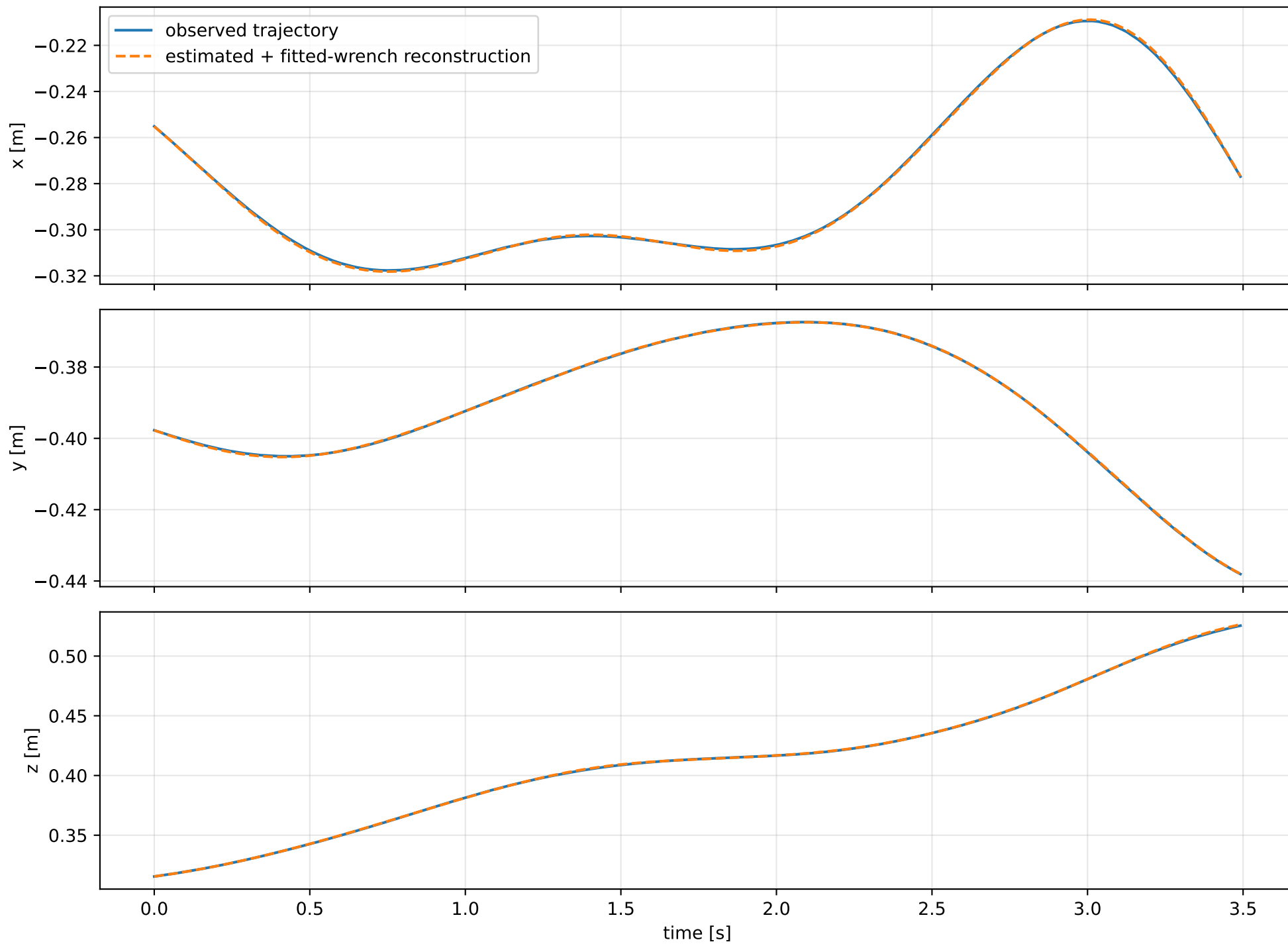
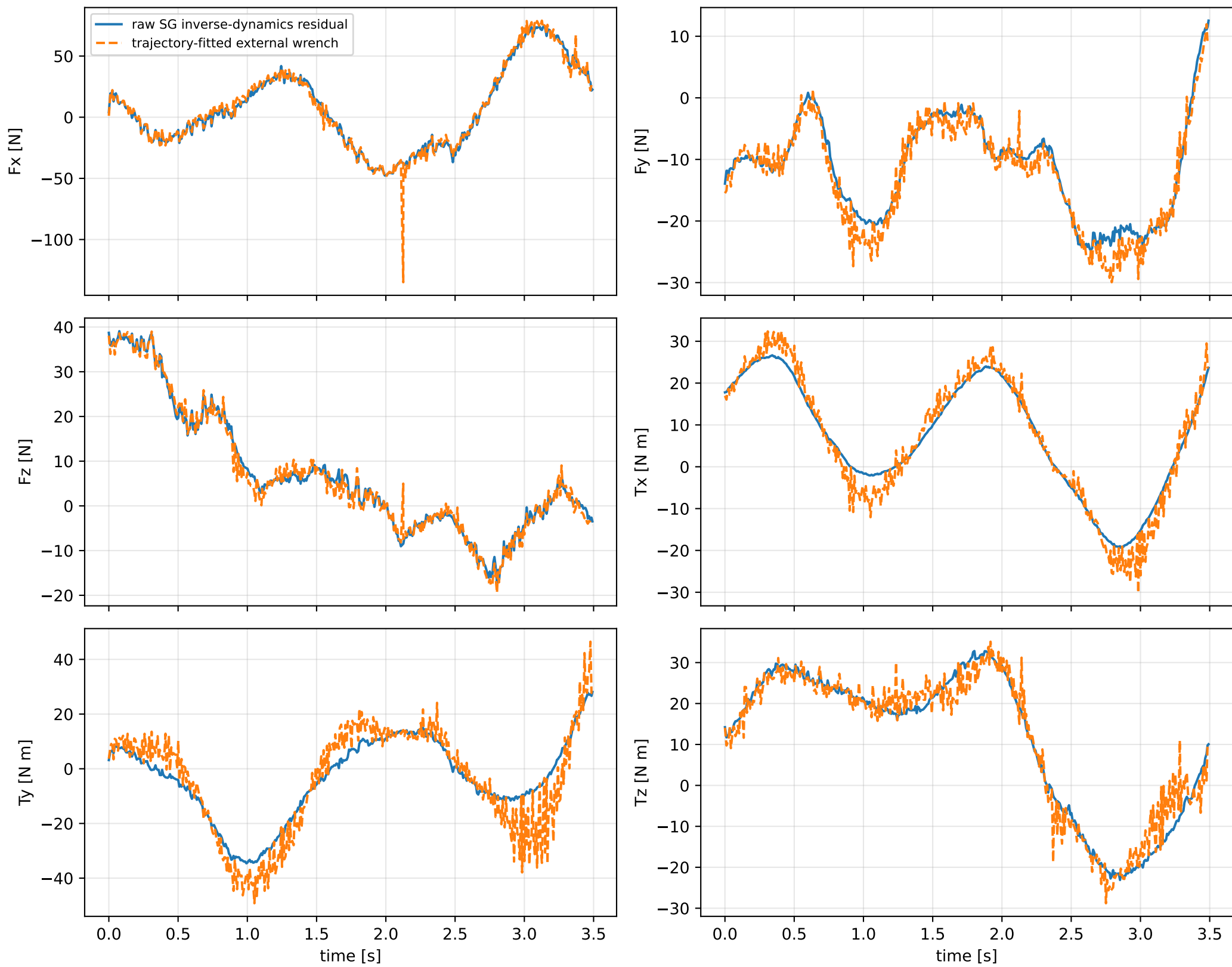


sweep_sg_window_covariance_004: trajectory

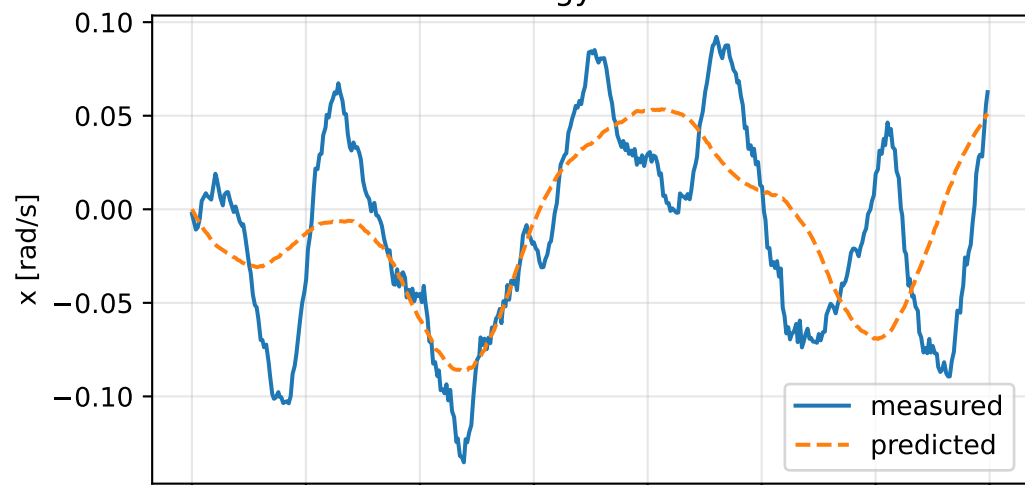


sweep_sg_window_covariance_004: residual wrench

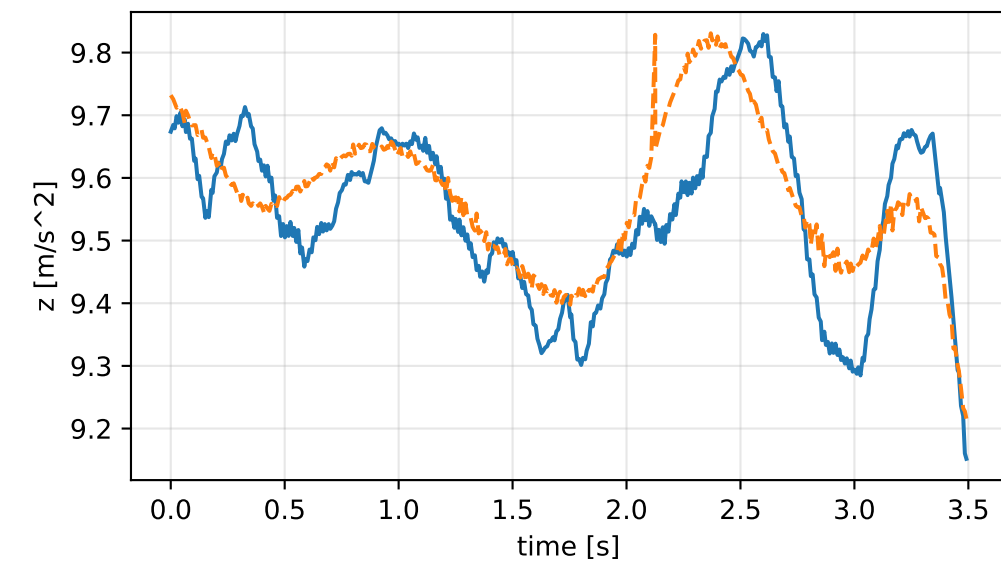
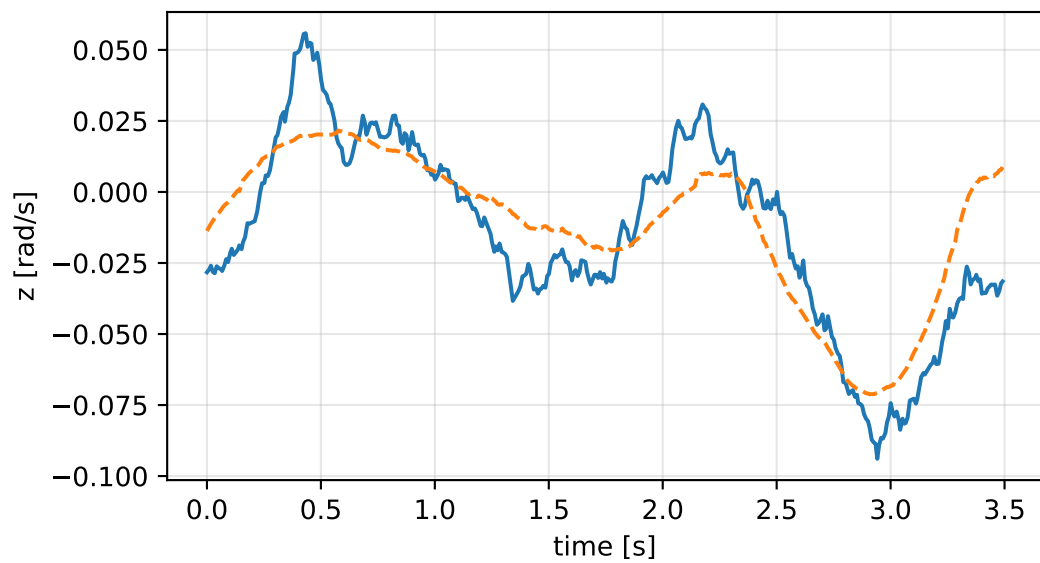
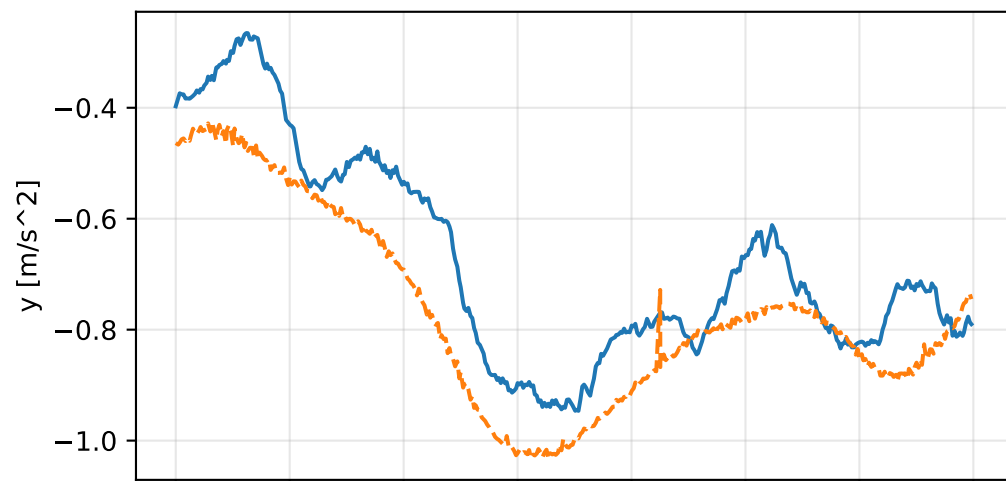
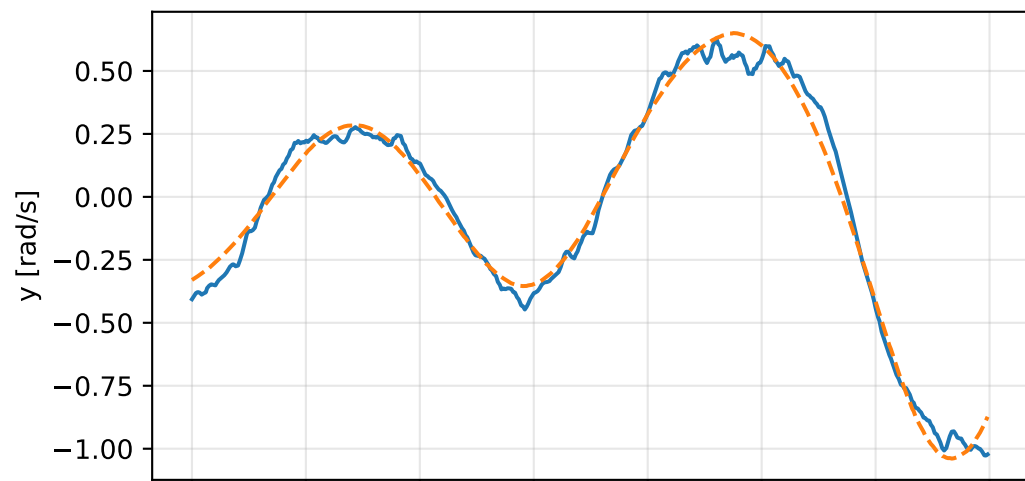
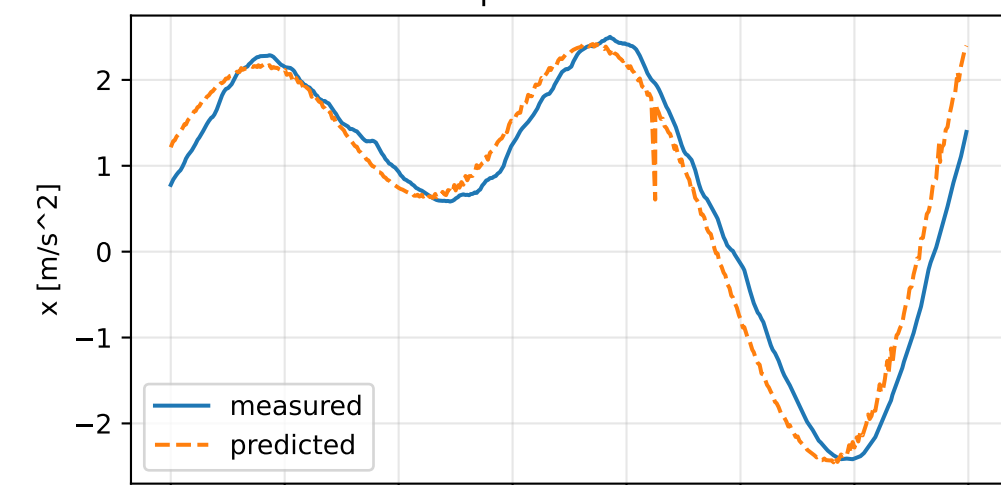


sweep_sg_window_covariance_004: measured sensor comparison

gyro



specific force



sweep_sg_window_covariance__004: nominal -> estimated parameters

Case: sweep_sg_window_covariance__004
status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 89.4695119 delta 87.1179143

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [22.436083 10.9925838 13.5822303] d=[22.3710829 10.9925845 13.5822113]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [10.9925838 32.039221 -8.9076999] d=[10.9925845 31.9742683 -8.9077]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [13.5822303 -8.9076999 37.2407212] d=[13.5822113 -8.9077 37.1117291]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-0.015707 0.24698 -0.2163533] d=[-0.0136823 0.2470105]

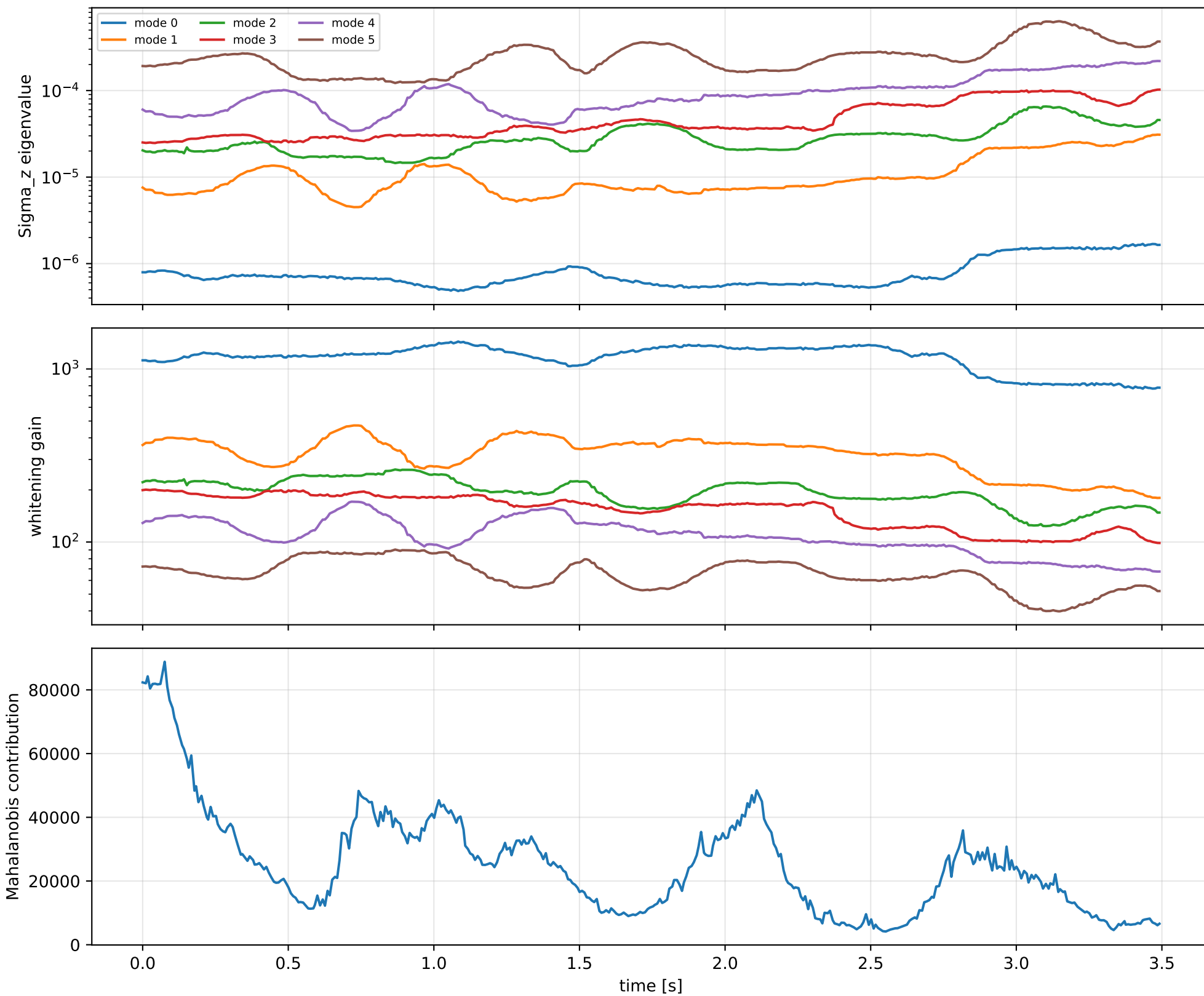
rotor force effectiveness

[1. 1. 1. 1.] -> [2.7953634e-05 9.2089502e-05 6.4698128e+01 6.2889687e+01] d=[-0.999972 -0.9999079 63.6981283 61.8896865]

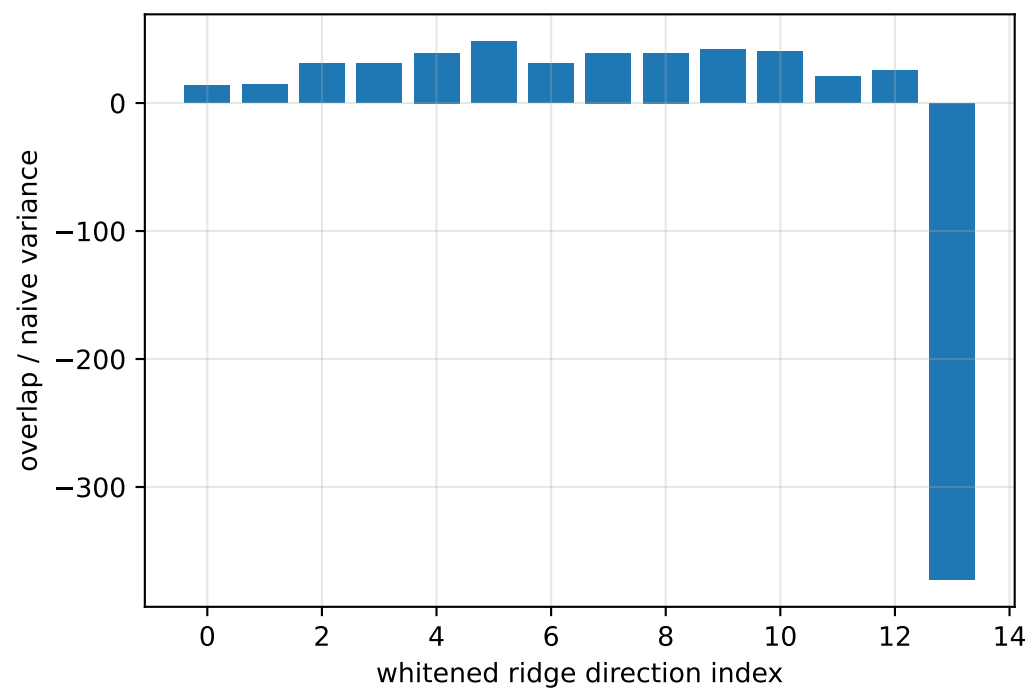
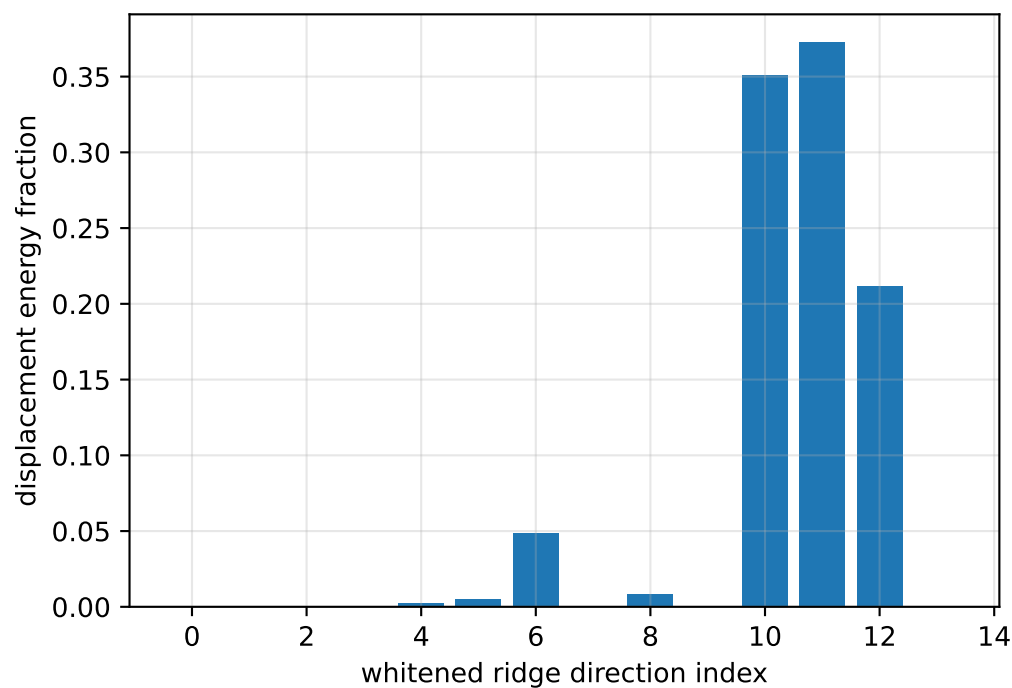
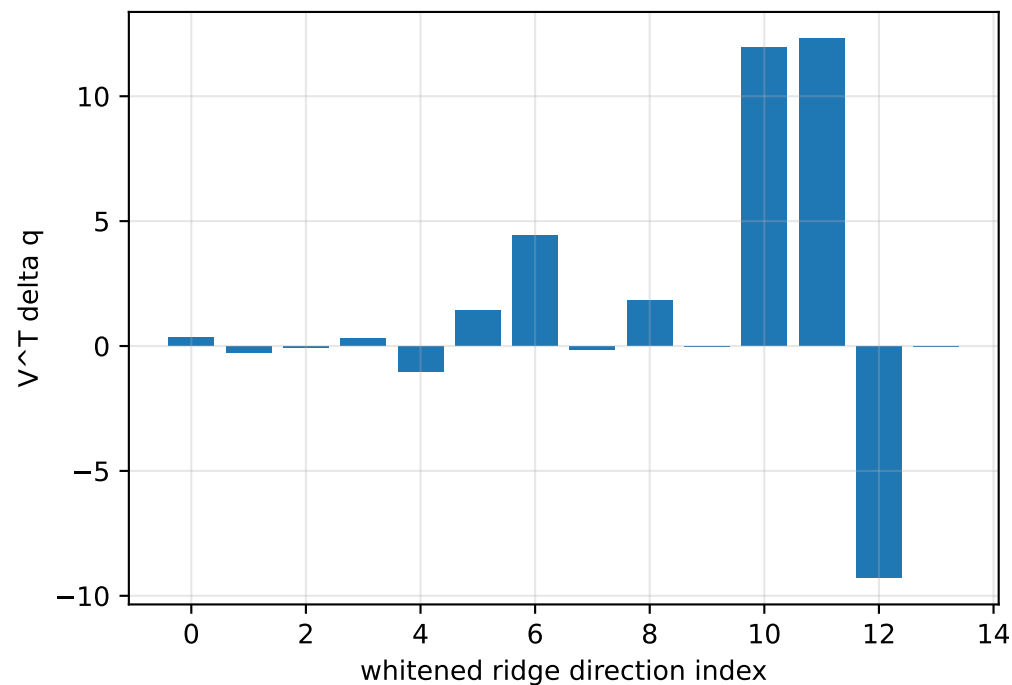
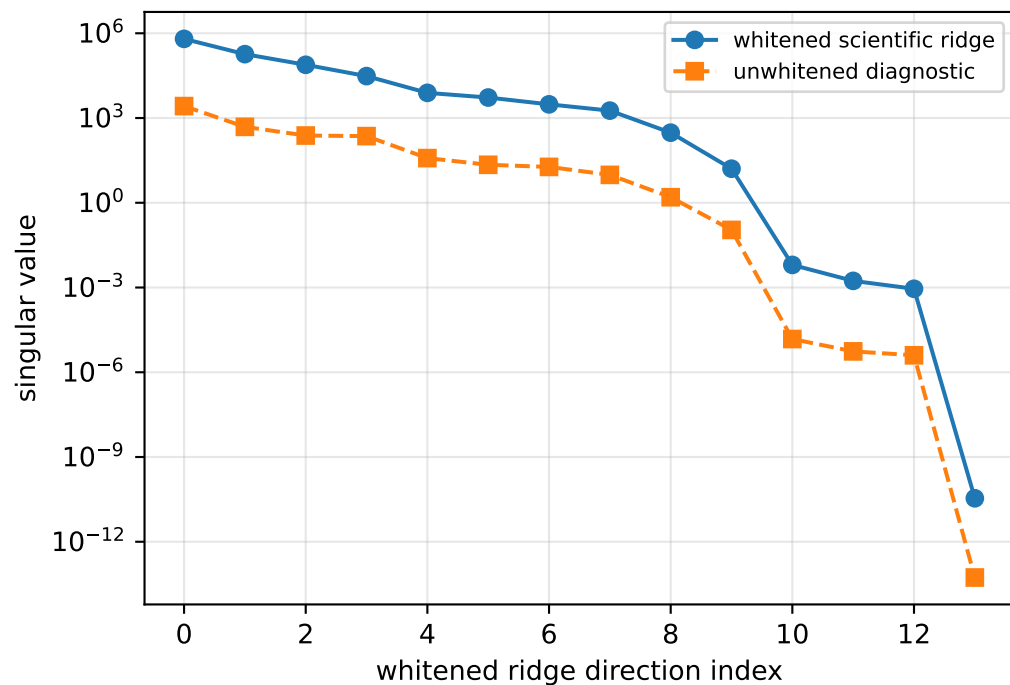
rotor lag [s] 0.2

gimbal lag [s] 0.0842723846

sweep_sg_window_covariance_004: covariance weighting

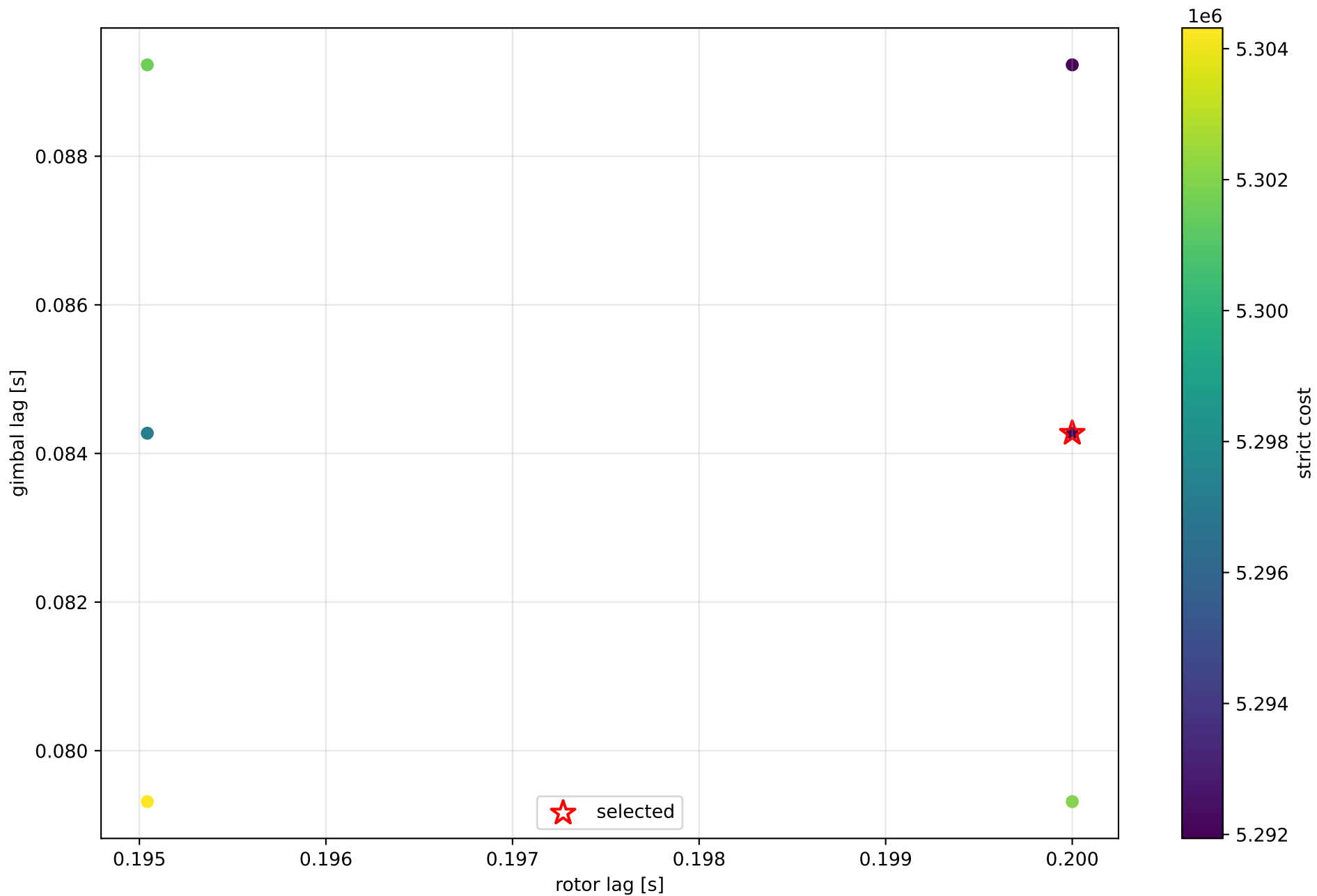


sweep_sg_window_covariance_004: parameter ridge



exact scale gauge: || v_{scale} ||=4.16889e-11, rank=13, nullity=1

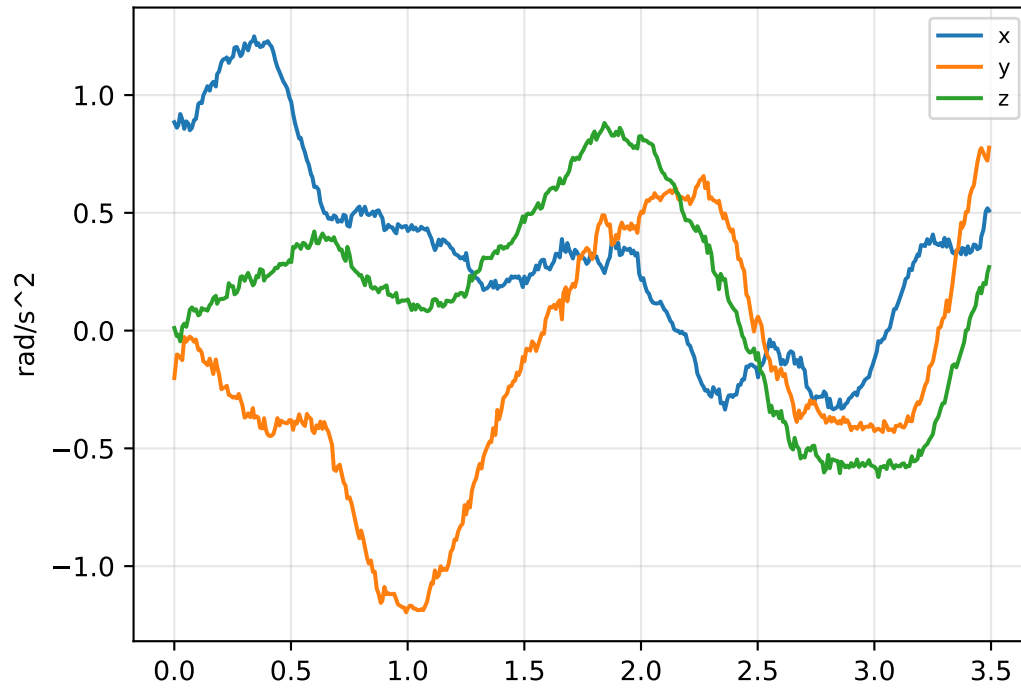
sweep_sg_window_covariance__004: strict lag candidates



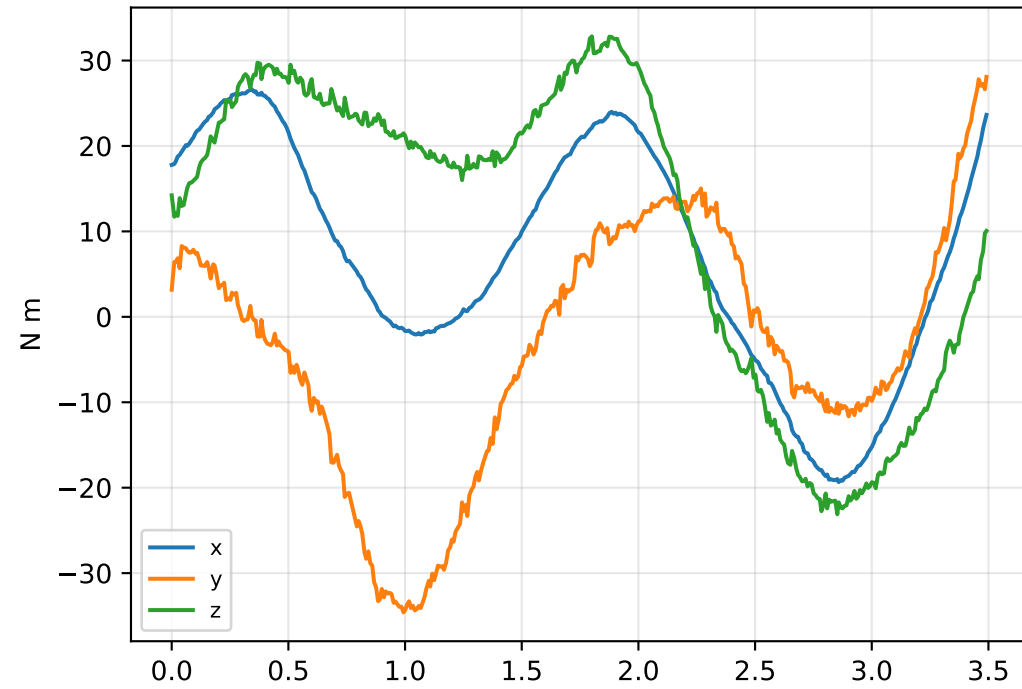
boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False

sweep_sg_window_covariance_004: acceleration/wrench closure

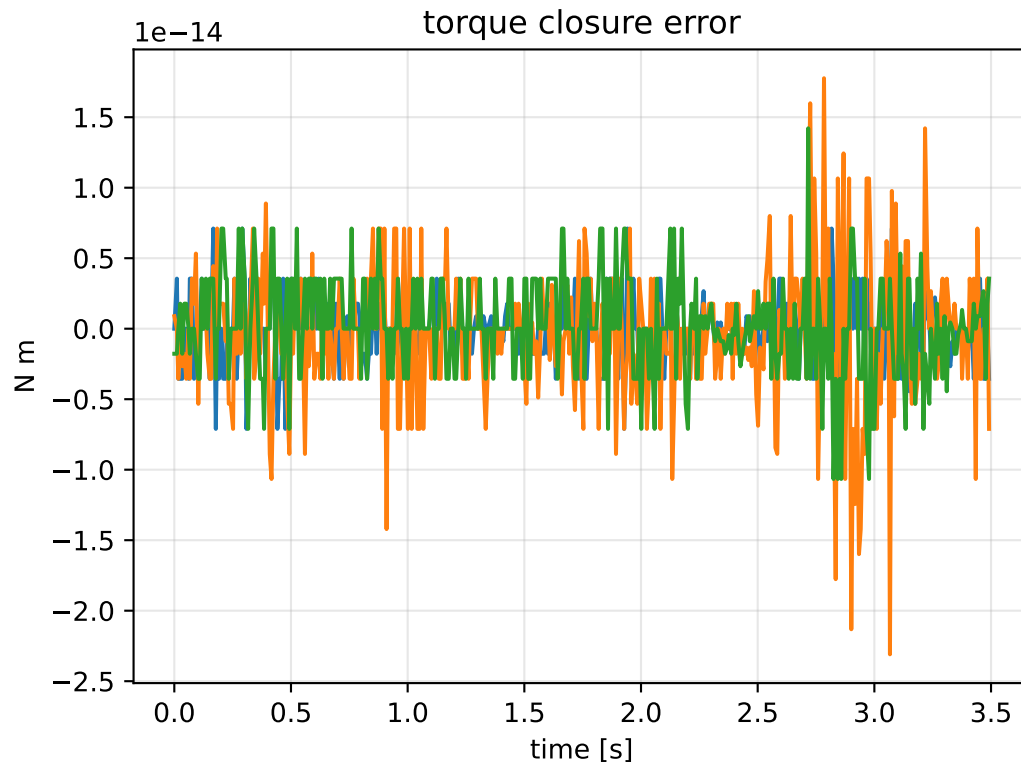
angular-acceleration residual



raw torque residual



torque closure error



principal moments [kg m²]
 nominal: [0.064953 0.065 0.128992]
 estimated: [6.62491 39.233122 45.857993]
 estimated / nominal: [101.996006 603.585863 355.510047]

force closure max/rms: 2.15827e-13 / 5.04561e-14
 torque closure max/rms: 2.30926e-14 / 3.58809e-15